

Beyond Accuracy: A Survival-Based Evaluation Framework for Elimination Reasoning in Medical Multiple-Choice Question Answering

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Abstract

Standard accuracy metrics evaluate whether a model selects the correct final answer in medical multiple-choice question answering (MCQA), but reveal nothing about how the model navigates the reasoning process that leads to that answer. This limitation is especially critical in clinical settings, where eliminating the correct diagnosis early constitutes a catastrophic and potentially irreversible failure. In this paper, we introduce a Survival-Based Evaluation Framework that tracks the correct answer’s survival across sequential elimination rounds rather than measuring only final accuracy. We further propose Confidence-Guided Elimination (CGE), a novel iterative elimination procedure in which the model assigns simultaneous numerical confidence scores to all options, and the lowest-scored option is eliminated each round. Experiments on the full MedMCQA validation split (4,183 questions) using Qwen2.5-7B-Instruct and LLaMA-3.1-8B-Instruct demonstrate that CGE significantly outperforms direct prediction for both models (+7.2% and +11.4% respectively). Surprisingly, augmenting CGE with few-shot examples provides no additional benefit, slightly hurting performance for both models. A subsequent Chain-of-Thought investigation on targeted failure cases reveals that catastrophic elimination failures are not caused by medical knowledge gaps but by verbalization gap. Models cannot accurately translate their internal confidence into numerical scores, a finding confirmed by logit analysis showing internal probability spreads of 0.791 (Qwen) and 0.379 (LLaMA) on cases where verbalized scores were flat. Our framework provides a diagnostic tool for model evaluation prior to deployment, revealing reasoning and reliability properties invisible to standard accuracy metrics.

1 Introduction

Medical multiple-choice question answering (MCQA) is a demanding evaluation setting for

large language models (LLMs). Questions from high-stakes clinical examinations contain carefully designed distractors, and human test-takers often solve them through elimination. Because each decision depends on the remaining options, elimination forms a natural sequential reasoning process. Whether LLMs can replicate this behavior, and whether current metrics can detect it, remains unclear.

Despite impressive benchmark performance (Hendrycks et al., 2020; Achiam et al., 2023), current LLM evaluation in MCQA collapses performance to a single scalar: final accuracy. This metric is blind to the reasoning path that produced the answer. Two models with identical accuracy can reason in completely different ways: one may carefully eliminate wrong options at every step while another arrives at the correct answer purely by luck. Final accuracy cannot distinguish between these two behaviors, yet they are not equivalent in clinical decision support. A system that rules out the correct diagnosis in the first elimination round is dangerous in ways that final accuracy simply cannot capture. The correct treatment option has been discarded before any further reasoning can recover it. Prior work on elimination-based reasoning has focused primarily on large models and general-domain benchmarks. Zhao and Zhang (2025) demonstrated consistent accuracy gains using Chain-of-Thought (CoT) elimination with Qwen-7B on educational datasets. Ma and Du (2023) proposed Process of Elimination (POE) effective at 7B+ scale, and Balepur et al. (2024) showed that LLMs inherently struggle with rejection-type reasoning due to a fundamental tension with instruction-tuning objectives.

Our contributions are:

- **A Survival-Based Evaluation Framework** with five new metrics: Survival@3, Survival@2, Accuracy@1, Catastrophic Failure

Rate (CFR), and Confidence-Survival Correlation. Using these we characterize the full elimination trajectory rather than only the final answer. We treat elimination as a time-to-failure process, drawing methodological inspiration from survival-style evaluations of sequential LLM robustness (Li et al., 2025).

- **CGE (Confidence Guided Elimination)**, a novel iterative elimination procedure in which the model assigns simultaneous 0 - 10 numerical confidence scores to all options, with the lowest scored option getting eliminated in each round.
- **The first study applying iterative Confidence Guided Elimination to MedMCQA** (4,183 questions), demonstrating CGE consistently outperforms direct prediction for both Qwen-7B and LLaMA-8B.
- **CoT and logit investigations** on targeted failure cases revealing a verbalization gap as the root cause of catastrophic elimination failures. Models possess the medical knowledge internally but cannot accurately express it as numerical confidence scores.

Importantly, we propose CGE and the Survival Framework not as a replacement for standard direct inference, but as a diagnostic evaluation tool intended to be run once on a model before deployment to stress-test its reasoning reliability and reveal failure modes invisible to accuracy alone. Just as engineers crash-test vehicles before they reach consumers, our framework evaluates a model’s elimination behavior before it is trusted in clinical settings.

We ask two research questions: **(RQ1)** Can LLMs perform meaningful elimination reasoning on medical MCQ, protecting the correct answer across sequential rounds beyond what chance predicts? **(RQ2)** Can survival-based metrics reveal reliability properties that final accuracy alone conceals, and if so, what is the root cause of elimination failures?

2 Related Work

2.1 LLMs for Medical Question Answering

Early MCQA systems relied on fine-tuned encoder models such as BERT (Devlin et al., 2019) on domain-specific corpora. Pal et al. (2022) introduced MedMCQA from Indian medical entrance

examinations covering 21 clinical subjects, providing a large-scale benchmark for medical NLP. Despite these advances, evaluation has remained accuracy-focused, obscuring failure modes that are clinically significant. Qin et al. (2024) demonstrated that LLMs exhibit systematic miscalibration on atypical medical presentations, motivating deeper reliability evaluation beyond accuracy.

2.2 Chain-of-Thought and Prompting Strategies

Wei et al. (2022) showed that prompting LLMs with intermediate reasoning steps substantially improves performance on complex reasoning. Kojima et al. (2022) extended this to zero-shot settings with the instruction “Let’s think step by step.” Wang et al. (2022) introduced self-consistency decoding, sampling multiple reasoning paths and selecting the most frequent answer. Weng et al. (2022) demonstrated self-verification through iterative reasoning passes. Our CGE strategy replaces verbal reasoning with structured numerical scoring, providing a more robust and parseable signal.

2.3 Elimination-Based Reasoning

Ma and Du (2023) introduced POE (Process of Elimination), a strategy that (1) scores all options, (2) removes those that score below average, (3) re-scores the remaining options, and (4) takes highest scored as the final answer. Balepur et al. (2024) demonstrated that elimination is fundamentally harder than selection for LLMs, as identifying incorrect options conflicts with instruction-tuning objectives. Zhao and Zhang (2025) proposed EBR integrating CoT elimination with Qwen-7B on educational benchmarks, explicitly calling for future work on medical domains and scale effects. Zhu et al. (2025) proposed Option-ID elimination using token-level probabilities for sequential option removal. None of these works apply elimination to MedMCQA, measure round-by-round survival, or investigate the verbalization gap.

2.4 Confidence Calibration in LLMs

Guo et al. (2017) established foundational calibration measurement for neural networks, showing systematic overconfidence. Kadavath et al. (2022) showed large LLMs can estimate their own uncertainty through direct probability elicitation. Tian et al. (2023) demonstrated that calibrated confidence can be elicited via explicit prompting strategies, even from RLHF-tuned models whose raw

likelihoods are distorted. Xiong et al. (2024) further showed that verbalized confidence scores are highly sensitive to prompt and model choice, directly grounding the verbalization gap risk we identify in our framework. Jain and Wallace (2019) demonstrated that attention weights do not reliably reflect model predictions, motivating explicit confidence measures. Our paper structure, methodology framing, and evaluation design follow three ACL-venue works as reference formats: Balepur et al. (2024) (ACL 2024 Findings) for elimination evaluation design, Ma and Du (2023) (EMNLP 2023) for strategy comparison, and Qin et al. (2024) (EMNLP 2024 Findings) for medical LLM reliability analysis.

3 Methodology

3.1 Dataset and Task Definition

We use MedMCQA (Pal et al., 2022), collected from Indian medical entrance examinations (AIIMS PG and INICET) covering 21 clinical subjects. Each instance consists of a clinical question q and four answer options $O = \{A, B, C, D\}$ with a single correct answer $o^* \in O$. We evaluate on the **full validation split of 4,183 questions** (SEED=42). The task is inherently sequential: each elimination round depends on the surviving option set from the previous round, directly satisfying the requirement for sequential language data processing.

3.2 Data Preprocessing

All questions are tokenized using each model’s native tokenizer (maximum 512 tokens). Questions exceeding this limit are truncated from the end of the question body, preserving all answer options. Null byte characters causing tokenization errors were removed. For CGE rounds, the option set is dynamically updated after each elimination. Greedy decoding (do_sample=False) ensures deterministic, fully reproducible outputs. When verbalized scores cannot be parsed from model output, a uniform fallback score is assigned to prevent evaluation failure.

3.3 Model Selection

Qwen2.5-7B-Instruct (Team, 2024): A 7.6B parameter model (float16, 15.2 GB VRAM). Selected for strong instruction following and competitive medical knowledge at this scale.

LLaMA-3.1-8B-Instruct (Grattafiori et al., 2024): An 8.0B parameter model (float16, 16.0 GB

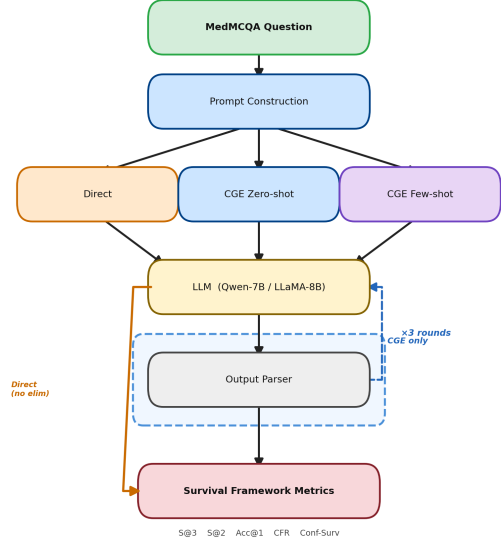


Figure 1: System pipeline. One-Shot makes a single inference call producing a direct answer. CGE loops the model–parser pair three times, eliminating the lowest-scored option each round.

VRAM). The most widely adopted open-source LLM family in NLP research, serving as a strong baseline.

Both models are evaluated without any fine-tuning on MedMCQA, measuring pre-trained medical knowledge under all three inference conditions: One-Shot, CGE Zero-shot, and CGE Few-shot.

3.4 System Pipeline

Figure 1 illustrates the complete evaluation pipeline. Three inference strategies are evaluated:

Direct: A single prompt with all four options. The model outputs one letter, serving as the standard MCQA baseline.

CGE Zero-shot (Novel): The model assigns integer scores 0 - 10 to all options simultaneously. Our parser eliminates the lowest-scored option, repeating for three rounds:

$$O_{t+1} = O_t \setminus \{\arg \min_{o_i \in O_t} s_t(i)\} \quad (1)$$

where $s_t(i)$ is the model’s verbalized score for option i at round t .

CGE Few-shot: Identical to CGE Zero-shot with three medical domain examples prepended, testing whether few sample demonstrations improves the numerical scoring quality.

3.5 Compute and Feasibility

All experiments use Google Colab Pro (A100, 40 GB VRAM). Total runtime is approximately

4 - 5 hours per model for 4,183 questions under greedy decoding. Checkpointing every 500 questions guards against session disconnection.

4 Evaluation Framework

We propose five survival based metrics characterizing the full elimination trajectory.

Survival@3 measures whether the correct answer survives after Round 1 (i.e., when 3 options remain). Random baseline: 75% - a model below this threshold performs worse than random at protecting the correct answer early.

Survival@2 measures whether the correct answer survives through Round 2 (i.e., when 2 options remain). Random baseline: 50%.

Accuracy@1 is equivalent to standard MCQA accuracy: the correct answer is the sole surviving option after all three elimination rounds. Random baseline: 25%.

Catastrophic Failure Rate (CFR) is the proportion of questions where the correct answer is eliminated in Round 1. Random baseline: 25%. A model below this threshold actively avoids the most dangerous failure mode in clinical AI.

Confidence-Survival Correlation measures for the questions where the model assigns the correct answer rank 1 in Round 1, what proportion end with the correct final answer. This directly quantifies whether expressed confidence is trustworthy.

5 Experiments and Results

5.1 Experimental Setup

All experiments use the MedMCQA full validation split (4,183 questions, SEED=42) with greedy decoding (do_sample=False) for fully deterministic outputs. Three inference strategies are evaluated: One-Shot, CGE Zero-shot, and CGE Few-shot. Only the inference strategy varies across experiments; all other factors like dataset, models, decoding configuration, and evaluation metrics are held constant to isolate strategy effects.

5.2 Experiment 1: Baseline Analysis

Table 1 shows both 7 - 8B models substantially exceed random chance (25%): Qwen-7B achieves 34.4% and LLaMA-8B 32.5% accuracy. This confirms genuine medical domain knowledge from pre-training (Hendrycks et al., 2020). Qwen-7B slightly outperforms LLaMA-8B on direct prediction despite fewer parameters, suggesting better

Model	Strat.	S@3	S@2	Acc	CFR	CS
Random	Base	75.0	50.0	25.0	25.0	—
Qwen-7B	Direct	—	—	34.4	—	—
	CGE-ZS	81.1	63.6	41.6	18.9	73.1
	CGE-FS	78.5	63.1	41.0	21.5	—
LLaMA-8B	Direct	—	—	32.5	—	—
	CGE-ZS	78.8	62.7	43.9	21.2	69.1
	CGE-FS	78.8	62.4	42.0	21.2	—

Table 1: Main results on MedMCQA full validation (4,183 questions). S@3/S@2=Survival@3/2, Acc=Accuracy@1, CFR=Catastrophic Failure Rate, CS=Confidence-Survival Correlation (all %). Bold = best per model.

Outcome	Qwen-7B	LLaMA-8B
Catastrophic (R1)	18.9% (792)	21.2% (885)
Late Failure	39.5% (1653)	35.0% (1462)
Success	41.5% (1738)	43.9% (1836)

Table 2: Failure distribution under CGE Zero-shot. Late failure dominates: models protect the correct answer through Round 1 but fail at the final two-option stage.

alignment with Indian medical examination content in MedMCQA.

5.3 Experiment 2: CGE Strategy Comparison

CGE significantly outperforms direct prediction.

Table 1 shows CGE Zero-shot raises Qwen’s accuracy from 34.4% to 41.6% (+7.2 pp) and LLaMA’s from 32.5% to 43.9% (+11.4 pp). Both models achieve Survival@3 above 75% random baseline (81.1%, 78.8%), and CFR below 25% random (18.9%, 21.2%), confirming active protection of the correct answer and avoidance of catastrophic failures.

Few-shot does not improve CGE. Adding three examples slightly hurts both models (Qwen: 41.6% → 41.0%; LLaMA: 43.9% → 42.0%). Few-shot raises Qwen’s CFR from 18.9% to 21.5%, suggesting pattern matching on example choices. LLaMA’s CFR is completely unchanged (21.2%).

Qwen has stronger confidence calibration. Qwen’s Confidence Survival Correlation is 73.1% vs. LLaMA’s 69.1%.

Error Analysis 1 - Question Length: Table 3 shows length has negligible effect on accuracy (30.9–35.5%), ruling it out as a confounding factor.

Error Analysis 2 - Failure Distribution: Table 2 shows late failure is the dominant error mode. Both models navigate Round 1 above chance but fail at the final two-option stage, locating the bot-

Length	Qwen-7B	LLaMA-8B
Short (<10 w)	34.9% (1754)	33.4% (1754)
Medium (10–20 w)	33.2% (1685)	30.9% (1685)
Long (>20 w)	35.5% (744)	34.3% (744)

Table 3: Direct prediction accuracy by question length. Near-uniform performance confirms length is not a confounding factor.

Question Set	CGE	CoT	Δ
Qwen2.5-7B			
Catastrophic Failures	0.0%	36.7%	+36.7
Flat Confidence	33.3%	26.7%	−6.7
Model Disagreements	23.3%	13.3%	−10.0
LLaMA-3.1-8B			
Catastrophic Failures	0.0%	43.3%	+43.3
Flat Confidence	33.3%	30.0%	−3.3
Model Disagreements	40.0%	53.3%	+13.3

Table 4: CoT investigation results (n=30 per set). CoT recovers substantial accuracy on questions where CGE failed, proving knowledge exists but cannot be expressed as numerical scores.

tleneck in fine-grained discrimination between the two most plausible options.

Error Analysis 3 - Catastrophic Failure Examples: Qualitative analysis reveals two patterns: (1) “all except” questions where models confuse selection vs. rejection framing; and (2) calculation questions requiring procedural knowledge that models assign low confidence to.

5.4 Experiment 3: CoT Investigation

To determine whether CGE failures reflect medical knowledge gaps or an inability to express knowledge as numerical scores, we applied zero-shot Chain-of-Thought (CoT) prompting. We asked models to reason freely in natural language to three targeted question sets of 30 questions each: (1) catastrophic failure cases where CGE eliminated the correct answer in Round 1, (2) flat confidence cases where the model assigned near-identical scores to all options, and (3) disagreement cases where Qwen and LLaMA produced different final answers.

Table 4 shows the key finding: on catastrophic failure questions (CGE = 0% by definition), CoT achieves 36.7% for Qwen and 43.3% for LLaMA. The same questions, same models yield dramatically better results under free-form reasoning, proving failures are caused by the numerical scoring format rather than knowledge absence.

On flat confidence cases, CoT slightly hurts both

Model / Set	Int.	Verb.	Gap?
Qwen flat cases	0.791	0.52	Yes
Qwen random base	0.928	7.56	—
LLaMA flat cases	0.379	0.24	Yes
LLaMA random base	0.513	8.40	—

Table 5: Logit analysis (n=25 per set). Int.=internal softmax spread (max−min over $\{A, B, C, D\}$); Verb.=verbalized 0–10 score spread. Both models exceed the 0.2 verbalization-gap threshold, confirming discriminative internal distributions despite flat expressed scores.

models (−6.7%, −3.3%), suggesting extended deliberation is counterproductive when the model is genuinely uncertain. On disagreement cases, LLaMA benefits from CoT (+13.3%) while Qwen is hurt (−10.0%), indicating Qwen’s first instinct outperforms its extended reasoning.

5.5 Experiment 4: Logit Analysis

To confirm whether the verbalization gap is a text-output problem or reflects genuinely flat internal representations, we extracted raw softmax probabilities over the four answer tokens directly from each model’s weights, motivated by logit-lens diagnostics showing that latent predictions in transformers can be decoded from internal representations (Belrose et al., 2023).

Table 5 confirms a verbalization gap for both models. On flat CGE cases, Qwen’s internal spread is 0.791 and LLaMA’s is 0.379. Both are well above the 0.1 weight-level threshold, proving the model knows the answer but cannot express it numerically. Qwen’s higher spread directly explains its stronger calibration (73.1% vs. 69.1% Confidence-Survival Correlation).

References

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, and 1 others. 2023. [GPT-4 technical report](#). *Preprint*, arXiv:2303.08774.
- Nishant Balepur, Shramay Palta, and Rachel Rudinger. 2024. [It’s not easy being wrong: Large Language Models struggle with process of elimination reasoning](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 4239–4256. Association for Computational Linguistics.
- Nora Belrose, Zack Furman, Logan Smith, Danny Halawi, Tuomas Oikarinen, Hailey Wu, Boaz Barak, and Jacob Steinhardt. 2023. [Eliciting latent predictions from transformers with the tuned lens](#). *Preprint*, arXiv:2303.08112.

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4171–4186. Association for Computational Linguistics.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, and 1 others. 2024. [The Llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. 2017. [On calibration of modern neural networks](#). In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2020. [Measuring massive multitask language understanding](#). *Preprint*, arXiv:2009.03300.
- Sarthak Jain and Byron C Wallace. 2019. [Attention is not explanation](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3543–3556. Association for Computational Linguistics.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, and 1 others. 2022. [Language models \(mostly\) know what they know](#). *Preprint*, arXiv:2207.05221.
- Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. [Large language models are zero-shot reasoners](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 22199–22213.
- Yuheng Li and 1 others. 2025. [Time-to-inconsistency: A survival-analytic perspective on LLM robustness](#). *Preprint*, arXiv:2502.10718.
- Chenkai Ma and Xinya Du. 2023. [POE: Process of elimination for multiple choice reasoning](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7183–7199. Association for Computational Linguistics.
- Ankit Pal, Logesh Kumar Umapathi, and Malaikanan Sankarasubbu. 2022. [MedMCQA: A large-scale multi-subject multi-choice dataset for medical domain question answering](#). In *Proceedings of the Conference on Health, Inference, and Learning*, pages 248–260. PMLR.
- Yuxin Qin, Zhangyue Xu, Yun Ye, Tao Gui, Qi Zhang, and Xuanjing Huang. 2024. [Enhancing healthcare LLM trust with atypical presentations recalibration](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 2520–2537. Association for Computational Linguistics.
- Qwen Team. 2024. [Qwen2.5 technical report](#). *Preprint*, arXiv:2412.15115.
- Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D Manning. 2023. [Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback](#). *Preprint*, arXiv:2305.14975.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022. [Self-consistency improves chain of thought reasoning in language models](#). *Preprint*, arXiv:2203.11171.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, and 1 others. 2022. [Chain-of-thought prompting elicits reasoning in large language models](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837.
- Yixuan Weng, Minjun Zhu, Fangyu Xia, Bin Li, Shizhu He, Shengping Liu, Bin Sun, Kang Liu, and Jun Zhao. 2022. [Large language models are better reasoners with self-verification](#). *Preprint*, arXiv:2212.09561.
- Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. 2024. [Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs](#). *Preprint*, arXiv:2306.13063.
- Qianli Zhao and Mei Zhang. 2025. [Elimination-based reasoning with LLM for multiple-choice educational question answering](#). *Journal of King Saud University – Computer and Information Sciences*, 37:204.
- Zhenduo Zhu, Bing Liu, Qingyao Ai, and Yiqun Liu. 2025. [Option-ID based elimination for multiple choice questions](#). *Preprint*, arXiv:2501.15175.